

Kartikeyan Sundaresan

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SUMMARY

Backend Engineer / Java Software Engineer with **5+ years** building distributed, event-driven commerce systems at scale across **HCL Commerce / WebSphere Commerce** and microservices ecosystems, owning architecture, migration strategy, production resiliency, and business-critical platform modernization. Delivered systems at **99.9%+ availability** serving **100,000+ peak users** and **£1.2M** incremental revenue.

CORE COMPETENCIES

Distributed Systems | Event-Driven Architecture | Microservices Architecture | System Design | High Availability | Fault Tolerance | API Design | Domain Driven Design | Performance Optimisation | Production Debugging | HCL Commerce / WCS | Identity Systems | Checkout Systems | Transactional Systems | Data Modeling

EXPERIENCE

Tata Consultancy Services

Software Development Engineer — Backend (Java)

Chennai, India

Aug 2020 – Present

Order & Delivery Systems (2022 – Present)

- Designed and owned Order subsystems for a UK retail e-commerce platform built on **HCL Commerce (WCS)**, maintaining **99.9%+ availability** at **50,000 daily users** and **100,000+ peak users** during high-traffic events (Christmas, flash sales).
- Led design and delivery of **Split Shipment Delivery** service with stateless, horizontally scalable OMS handlers for dynamic item grouping and delivery date retrieval, driving a **10% increase in delivery revenue**; engineered **same-day Click & Collect** checkout flow generating **£1.2M incremental revenue**.
- Designed **Kafka async event pipeline** (Checkout Basket to Order Fulfillment) with **idempotent consumers and dead-letter queues**. Chose pub-sub over synchronous REST to decouple services and absorb 10x traffic spikes without cascading failures; led RCA and permanent remediation for critical production incidents.
- Conducted technical sessions for junior engineers on secure coding practices, design pattern implementation, and checkout performance optimisation, translating production incident learnings into team-wide engineering standards.
- Owned CI/CD pipelines via GitHub Actions and managed containerised deployments using Docker, monitored customer-facing order services via Dynatrace, triaging alerts on order failures and exception spikes.
- Extended **WCS backend controllers** to integrate the **UK Deposit Return Scheme**; enabled deposit collection, customer returns, and automated refunds for eligible products while managing deposit calculations, Oracle SQL persistence, and downstream payment and fulfillment integrations, with changes deployed via **Management Center** configuration and custom command extensions.
- Resolved **XSS, CSRF, IDOR** production vulnerabilities; achieved **97% unit test coverage** on legacy modules with JUnit and Mockito.

Accounts & Identity Systems (2020 – 2022)

- Drove end-to-end **CIAM migration**: designed and rewrote versioned, backward-compatible session establishment APIs consumed across **15+ downstream domain services**, migrating identity management from monolithic **WCS** to a microservices architecture with zero service disruption and zero rollback events.
- Implemented **OAuth 2.0 + JWT** authentication, choosing stateless tokens over server-side sessions to eliminate session affinity requirements, enabling free load-balancer routing across personalisation, payments, and fulfilment services.

PROJECTS

GondorGates—Distributed Rate-Limiting Sidecar Gateway

github.com/Kartik199/GondorGates

Spring WebFlux • Redis • Lua • Docker • Prometheus • Micrometer • Grafana

- Built a production-grade distributed rate-limiting sidecar gateway intercepting every request before it reaches the backend. Validated via **k6 load tests** at 100 concurrent VUs (**P95 ~14ms, avg ~7ms, ~449 req/s**); GondorGates overhead isolated to **~5ms** (one Redis Lua round-trip); atomicity confirmed at exactly 5/40 requests allowed under 20 concurrent VUs.
- **Reactive architecture**: Selected Spring WebFlux over Spring MVC for its non-blocking event-loop model; MVC's thread-per-request model exhausts its thread pool under high concurrency while WebFlux pipelines requests without blocking. Throughput ceiling deliberately shifted to Redis network I/O, demonstrating intentional hardware/software bottleneck placement.

- **Atomic multi-dimensional enforcement:** Designed a hierarchical resolver evaluating four dimensions (Global, User, IP, and API Key) with short-circuit on first denial; each read-refill-decide-write sequence executes atomically in a **single Redis Lua round-trip**, choosing Lua over application-level locking (latency penalty) and WATCH/MULTI/EXEC (retry storm risk) for O(1) enforcement with no race conditions.
- **Runtime resilience:** Built a Bearer-token-authenticated Admin API for runtime policy overrides with JSR-380 validation and a **DIP-compliant PolicyStore** interface; implemented **fail-open strategy** ensuring continued availability if Redis becomes unreachable.
- **Full observability:** Custom auto-provisioned Grafana dashboard tracking request rates, block rate, per-endpoint allow/deny, filter latency (P50/P95), Redis eval duration, and live token bucket levels per dimension; Redis fail-open alert rule included; **51 tests (unit + integration)** validated in CI via GitHub Actions.

Selvora — Personal Finance Tracker

github.com/Kartik199/selvora-api

Spring Boot • PostgreSQL • ReactJS • JWT

- Integrates **live NAV data** from external APIs with a **circuit-breaker and TTL cache** for production resiliency under provider outages, maintaining portfolio visibility without serving indefinitely stale data.
- Computes **XIRR** for investment portfolio returns, maps investments to user-defined financial goals, and models **recurring transaction approval flows** with async prompts gating ledger commits, mirroring real-world banking authorisation patterns.
- Implements multiple **EMI calculation strategies**, monthly cash flow projections, and spending analytics; secured with **JWT stateless auth** for horizontal scalability without session store dependencies.

TECHNICAL SKILLS

Backend Architecture:	Java (Primary), Spring Boot, Spring WebFlux (Reactive Streams)
Commerce Platforms:	HCL Commerce (WCS) v9, OMS, Checkout Systems, WCS Controllers, Management Center
Systems Design:	Apache Kafka, Redis, Microservices, REST APIs, Lua (Redis scripting)
Data Stores:	PostgreSQL, Oracle SQL, MongoDB
Observability:	Prometheus, Micrometer, Grafana, Dynatrace
Cloud & DevOps:	AWS (EC2, S3, IAM) • Docker • GitHub Actions
Security:	OAuth 2.0, JWT, CIAM, XSS / CSRF / IDOR remediation
Testing:	JUnit, Mockito

EDUCATION

Amrita Vishwa Vidyapeetham University
B.Tech in Computer Science and Engineering

2016 – 2020
GPA: 7.92 / 10.0